Nassau Candy Distributor

Product Line Profitability & Margin Performance Analysis

Project Type: Data Analytics

Tools: Python, Pandas, Plotly, Google Colab

1. Background and Context

Nassau Candy Distributor operates across multiple product lines and regions. Understanding product profitability, sales performance, and factory-wise contribution is important for effective business decisions.

This project analyzes the distributor’s sales dataset to identify profitable products, evaluate gross margins, and understand factory performance.

2. Problem Statement

The objective is to analyze sales and profitability data to:

* Identify the highest-profit-generating products.
* Evaluate regional and division-wise performance.
* Analyze monthly revenue trends.
* Compare factory-wise profitability.
* Identify products contributing significantly to overall profit through Pareto analysis.

3. Dataset Description

The dataset contains 10,194 records and 18 columns.

Important fields include:

* Order Date
* Ship Date
* Division
* Region
* Product Name
* Sales
* Units
* Gross Profit
* Cost

4. Analytical Methodology

The project followed these steps:

1. Imported the dataset into Google Colab.
2. Cleaned column names and converted date columns.
3. Removed duplicate records.
4. Calculated business KPIs.
5. Created product-wise and region-wise visualizations.
6. Analyzed monthly revenue trends.
7. Performed Pareto analysis.
8. Created factory-wise profitability analysis.
9. Developed interactive Plotly visualizations.

5. Key Performance Indicators

KPI	Result
Total Revenue	141,783.63
Total Gross Profit	93,442.80

Total Units Sold	38,654
Overall Gross Margin	65.90%
Records Analyzed	10,194

6. Product Profitability Analysis

The product-wise analysis identifies products generating the highest gross profit.

The highest-profit-generating product in the analysis was:

Wonka Bar - Scrumdiddlyumptious

Gross Profit: 19,357.50

7. Factory-wise Performance Analysis

Factory	Revenue	Gross Margin
Lot's O' Nuts	76,340.15	69.13%
Wicked Choccys	55,352.75	65.13%
Secret Factory	8,587.50	50.59%
Sugar Shack	220.98	54.86%
The Other Factory	1,282.25	11.87%

Lot's O' Nuts recorded the highest revenue and gross margin among the five mapped factories.

The Other Factory recorded the lowest gross margin, suggesting an area for further investigation.

8. Pareto Analysis

Pareto analysis was performed using product-wise gross profit.

Products were ranked in descending order of gross profit, and cumulative profit percentages were calculated to understand how much of the total profit is generated by the leading products.

9. Factory Coordinates and Product-Factory Correlation

Factory coordinates were used to create a geographic visualization of factory locations.

Product-to-factory mapping was used to analyze gross profit, sales, and units across factories.

All 10,194 records were mapped in the final notebook output.

10. Conclusion

The analysis provides an overview of product profitability, factory contribution, and overall business performance.

Key findings include:

* Total revenue of 141,783.63.
* Total gross profit of 93,442.80.
* Overall gross margin of approximately 65.90%.
* Lot’s O’ Nuts was the leading factory by revenue.
* Wonka Bar - Scrumdiddlyumptious was the leading product by gross profit.

The interactive dashboards and analytical results can support business performance monitoring and better decision-making.

11. Tools and Technologies

* Python
* Pandas
* Plotly

* Google Colab

* Streamlit (planned separately; application deployment pending)

12. Project Deliverables

* Python analysis notebook (.ipynb)

* Interactive Plotly dashboard (HTML)

* Factory profitability analysis (HTML)

* Factory location map (HTML)

* Executive summary and project report

Note: Factory mapping and coordinates should be checked against the official project brief before final submission. The Streamlit application is not yet completed, and the Plotly HTML dashboard does not replace that requirement.